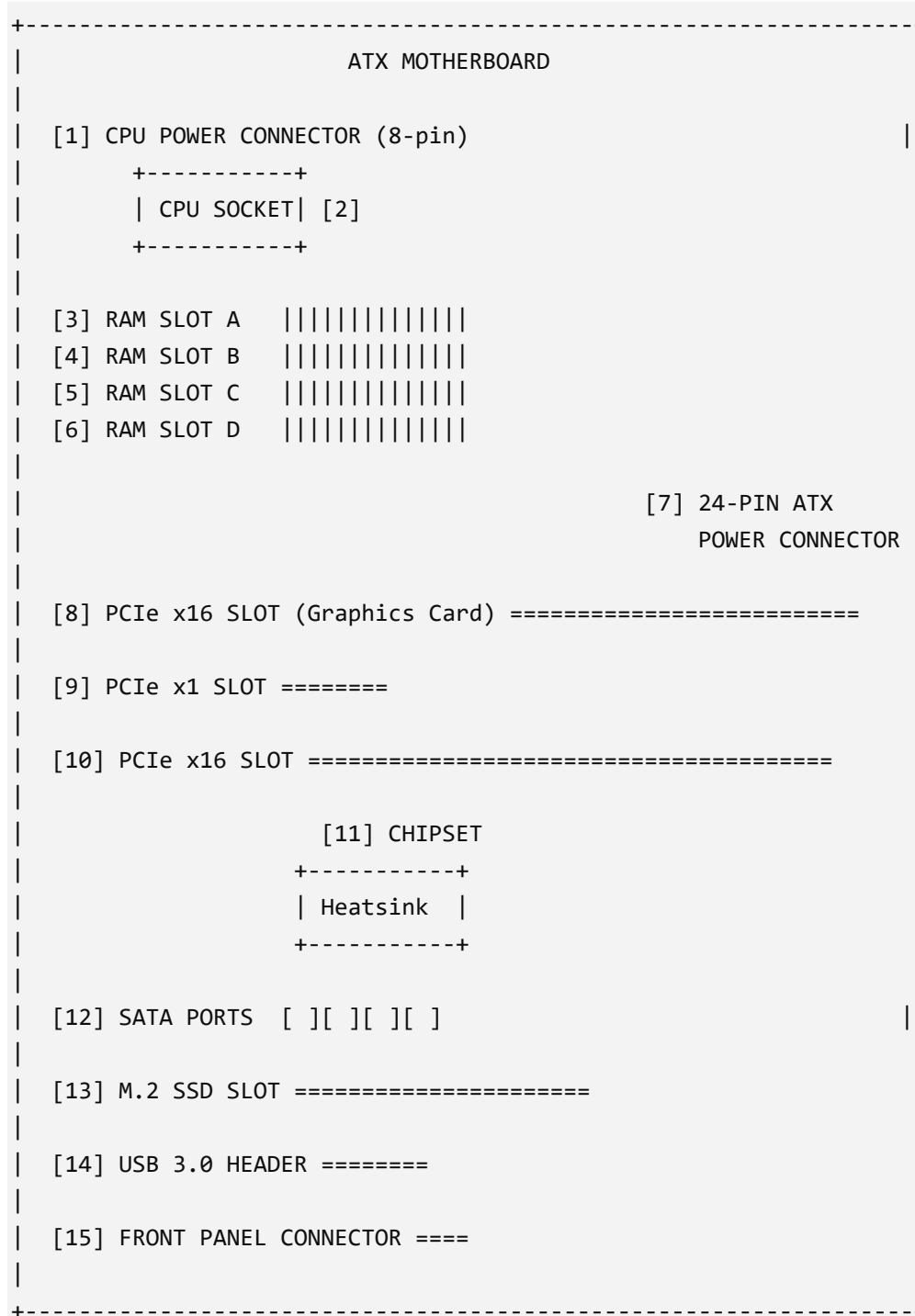


Motherboard and CPU Assignment

1. Labeled Diagram of an ATX Motherboard



Components Marked

1. CPU Power Connector
2. CPU Socket
3. RAM Slot A
4. RAM Slot B
5. RAM Slot C
6. RAM Slot D
7. 24-Pin ATX Power Connector
8. PCIe x16 Slot
9. PCIe x1 Slot
10. PCIe x16 Slot (secondary)
11. Chipset
12. SATA Ports
13. M.2 SSD Slot
14. USB 3.0 Header
15. Front Panel Connector

2. Popular CPU Socket Types and Compatible CPUs

CPU Socket Type	Manufacturer	Example Compatible CPU
LGA1151	Intel	Intel Core i7-9700K
AM4	AMD	AMD Ryzen 5 5600X
LGA1200	Intel	Intel Core i5-10400

Notes

- LGA1151 supports many Intel 6th–9th generation processors.
- AM4 supports multiple Ryzen generations.
- LGA1200 supports Intel 10th and 11th generation desktop processors.

3. Functions of the CPU and Chipset

CPU (Central Processing Unit)

The CPU is the main processor of the computer. It executes instructions, performs calculations, manages programs, and controls most operations. It is often called the "brain" of the computer because it processes data and makes decisions.

Chipset

The chipset is a group of controller chips on the motherboard. It manages communication between the CPU and other components such as USB devices, storage drives, network adapters, and expansion cards. The chipset helps organize and direct data traffic throughout the system.

How They Work Together When Opening Spotify

1. You click the Spotify icon.
2. The CPU receives the instruction and starts the application.
3. The chipset helps retrieve Spotify files from the SSD or HDD.
4. The CPU processes the program code and loads it into RAM.
5. The chipset manages communication with audio hardware, USB devices, and network connections.
6. Spotify connects to the internet and begins streaming music while the CPU continues processing commands and the chipset handles data transfer between components.

4. Real Motherboard Photo Identification Activity

